

Summer 2025, 4100 Problem Set 1

Please complete all problems. For any proof questions, please structure your proof similar to those from lecture.

1. Let $\mathbf{x}, \mathbf{y} \in \mathbb{R}^n$. Show that $\mathbf{x}^T \mathbf{y} = 0$ if and only if $\|\mathbf{x}\| \leq \|\mathbf{x} + a\mathbf{y}\|$ for all $a \in \mathbb{R}$.
2. Let $\{\mathbf{x}_1, \dots, \mathbf{x}_n\}$ be a set of vectors in $\mathbb{R}^n$ such that $\mathbf{x}_i^T \mathbf{x}_j = \begin{cases} 0 & i \neq j \\ 1 & i = j \end{cases}$. Show that if $\mathbf{v} = \sum c_i \mathbf{x}_i$ then $c_i = \mathbf{x}_i^T \mathbf{v}$. (Note that the vectors $\mathbf{x}_i$ are said to be orthonormal)
3. Let a , b , and c be lengths of the sides of a triangle and let d be the length of the line segment from the midpoint of the side of length c to the opposite vertex. Use inner products to prove that

$$a^2 + b^2 = \frac{1}{2}c^2 + 2d^2$$

(See figure on next page)

4. Let V be a vector space over F and let S and T be subspaces such that $S \cap T = \emptyset$. Show that for any $x \in S + T$, there exists unique $s \in S$ and $t \in T$ such that $x = s + t$. (Note that $\emptyset$ means just containing the zero vector)
5. For each of the following subsets of $\mathbb{R}^3$, either prove that they are a subspaces or show by counterexample that they are not a subspace.
 - a. $\{(x_1, x_2, x_3) \in \mathbb{R}^3 : x_1 = 4x_3\}$
 - b. $\{(x_1, x_2, x_3) \in \mathbb{R}^3 : x_1 x_2 x_3 \leq 0\}$
 - c. $\{(x_1, x_2, x_3) \in \mathbb{R}^3 : 2x_1 - x_2 + 3x_3 = 0\}$
 - d. $\{(x_1, x_2, x_3) \in \mathbb{R}^3 : x_1^2 - x_2^2 = 0\}$

